

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Electronics and Electrical Communication Engineering
2. **Subject Name:** Machine Learning and Artificial Intelligence for Electronic Systems

L-T-P: 3-1-0

Credits: 4

3. Pre-requisites:

Math pre-requisites -

LINEAR ALGEBRA, NUMERICAL AND COMPLEX ANALYSIS (MA11004)

PROBABILITY AND STOCHASTIC PROCESSES (EC1xxxx, new course from E&ECE)

LINEAR ALGEBRA AND OPTIMIZATION MODELS (EC21204)

E&ECE pre-requisites -

ELECTRONIC SYSTEMS (EC1xxxx, new course from E&ECE)

ANALOG ELECTRONIC CIRCUITS (EC21207)

DIGITAL ELECTRONIC CIRCUITS (EC21202)

SIGNALS AND SYSTEMS (EC21208)

ELECTROMAGNETIC ENGINEERING (EC1xxxx, changing from EC21206)

DIGITAL SIGNAL PROCESSING - I (EC2xxxx, changing from EC31201)

COMMUNICATION-I (EC2xxxx, changing from EC31202)

4. Syllabus and reference books:

Syllabus:

Introduction to Machine Learning, Regression and Classification, Neural Networks, Deep Learning, Neural Networks for Intelligent Sensors, Machine Learning for Spectral Analysis, Machine Learning for Electronic Design Automation, Machine Learning for Computational Electromagnetics, Deep Learning for Multidimensional Signal Processing, Neural Data Compression, Deep Learning for Time Series Analysis

Reference Books:

1. "Understanding Machine Learning – From Theory to Algorithms", Shai Shalev-Shwartz and Shai Ben-David, Cambridge University Press (2014)
2. "Pattern Recognition and Machine Learning", Christopher M. Bishop, Springer (2006)
3. "Deep Learning – Foundations and Concepts", Christopher M. Bishop and Hugh Bishop (2024)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Formal Introduction to Machine Learning: Motivation of Machine Learning, Types of Learning, Empirical Risk Minimization and General Learning Model, Model Training and Inference, Overfitting & Underfitting.	6
2.	Regression and Classification: Linear Models, Least Squares Methods, Logistic Regression, Naive Bayes, Perceptron, Support Vector Machine.	9
3.	Neural Networks: Multi-layer Perceptron, Backpropagation, Optimization, Regularization, Performance Evaluation.	6
4.	Deep Learning: Introduction to Deep Neural Networks, Convolutional Neural Networks, Deep Classification and Generation, Deep Learning Implementation Frameworks.	11
5.	Implementation in Electronic Systems and Applications: Machine Learning for Spectral Analysis Machine Learning for Electronic Design Automation Machine Learning for Computational Electromagnetics Neural Networks for Intelligent Sensors Deep Learning for Multidimensional Signal Processing Neural Data Compression Deep Learning for Time Series Analysis: Causal Inference and Security Threat Detection	12
Total number of hours		44